

# TRANSPORTATION TECHNOLOGY

Transportation affects our lives in a multitude of ways. Transportation systems move raw materials to manufacturers and finished products to consumers locally, nationally, and globally. Individuals use transportation systems every day for business, work, and pleasure.

Transportation technology courses provide students with opportunities to understand transportation systems from the perspective of either the consumer or the service provider. The range of courses enables students to study both vehicle ownership and vehicle maintenance, and to develop skills and prepare for careers in the servicing and repair of vehicles, aircraft, and/or watercraft. In addition, students will develop the Essential Skills and work habits that are important for success in the transportation industry.

The list of approved emphasis areas for transportation technology can be found at [www.edu.gov.on.ca/eng/curriculum/secondary/teched.html](http://www.edu.gov.on.ca/eng/curriculum/secondary/teched.html).

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

# Transportation Technology: Vehicle Ownership, Grade 11

Open

TTJ30

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This general interest course enables students to become familiar with the options and features of various vehicles, issues of registration, and the legal requirements affecting vehicle owners. Students will also learn about vehicle financing and insurance, vehicle maintenance, emergency procedures, and the responsibilities of being a vehicle owner. Students will develop an awareness of environmental and societal issues related to vehicle ownership and use, and will explore career opportunities in the transportation industry.

**Prerequisite:** None

# A. VEHICLE OWNERSHIP FUNDAMENTALS

## OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** assess factors affecting decisions on vehicle selection and purchase from an informed consumer's point of view;
- A2.** demonstrate an understanding of the vehicle registration process and the legal requirements of vehicle ownership;
- A3.** demonstrate an understanding of loans, leases, and insurance policies related to the purchase or lease of a vehicle;
- A4.** identify and describe the precautions, procedures, and obligations related to roadside emergencies and accidents.

## SPECIFIC EXPECTATIONS

### A1. Vehicle Selection

By the end of this course, students will:

- A1.1** identify various vehicle makes, models, and body styles and compare their features, options, and price;
- A1.2** outline the factors that determine the value of a used vehicle (*e.g., year, mileage, appearance, mechanical condition, warranties, vehicle history, options*);
- A1.3** identify the factors that an owner should consider when selecting the most appropriate vehicle to purchase or lease (*e.g., cost, size and type of vehicle, fuel economy [and government incentives, if applicable], ergonomics, intended type and frequency of vehicle use, time of year*) and explain their importance;
- A1.4** identify the reasons for "road testing" a vehicle before purchase (*e.g., to determine comfort and visibility, check for noise, test brakes and steering, assess performance and handling*);
- A1.5** describe on the basis of research various sales and marketing practices encountered in purchasing or leasing a vehicle (*e.g., high-pressure sales tactics, use of black or red book evaluations*);
- A1.6** describe potential issues involved in purchasing a vehicle from various sources (*e.g., dealerships, used car retail outlets, auctions, private sellers, Internet websites*).

### A2. Vehicle Registration and Ownership

By the end of this course, students will:

- A2.1** describe on the basis of research the legal requirements associated with owning and operating a vehicle (*e.g., fees, taxes, licensing, permits, proof of insurance*);
- A2.2** describe the conditions that must be met to register a vehicle (*e.g., safety standards inspection, emissions test, proof of insurance*);
- A2.3** explain the purpose of and the protection provided by the Used Vehicle Information Package.

### A3. Vehicle Loans and Insurance

By the end of this course, students will:

- A3.1** identify appropriate resources for use in the selection of a vehicle loan or lease and insurance policy;
- A3.2** explain vehicle insurance coverage requirements and options (*e.g., requirements: third-party liability coverage, statutory accident benefits coverage; options: collision or upset coverage, comprehensive coverage, loss of vehicle use coverage*);
- A3.3** explain the factors that affect the cost of vehicle insurance (*e.g., age and model of vehicle, age and gender of policy holder, driving record*);

**A3.4** compare the overall cost of loans from various sources (*e.g., banks, finance companies, private lenders*);

**A3.5** compare the advantages and disadvantages of purchasing versus leasing a vehicle (*e.g., with regard to short-term and long-term costs, ownership, and responsibility for maintenance and repairs*);

**A3.6** describe the criteria financial institutions use to determine eligibility for a loan or lease (*e.g., credit history, income, net worth*).

#### A4. Roadside Emergency Precautions and Procedures

By the end of this course, students will:

**A4.1** describe the legal consequences and obligations that arise when a driver is involved in an accident and/or charged with a traffic violation (*e.g., consequences: fine, charge, legal action; obligations: show proper documentation, remain at the scene of the accident*);

**A4.2** describe the steps to follow when involved in a traffic accident (*e.g., with respect to law enforcement involvement, exchanging insurance information, making a claim to an insurance company*);

**A4.3** compare various roadside assistance plans (*e.g., manufacturer's plan, aftermarket plan*);

**A4.4** identify safe procedures and recommended techniques (*e.g., for jacking, battery boosting, towing*) described in the owner's manual to resolve roadside emergencies (*e.g., flat tire, dead battery, vehicle breakdown*);

**A4.5** describe the preparations vehicle owners can make to increase driving safety (*e.g., trip planning, driver training, keeping the vehicle properly maintained*);

**A4.6** describe the precautions vehicle owners can take to minimize the effect of accidents or roadside emergencies (*e.g., make sure everyone wears a seatbelt; observe speed limits; have a first aid kit, emergency tool kit, and fire extinguisher in the vehicle; carry a cell phone; wear a helmet when riding a motorcycle*).

## B. VEHICLE MAINTENANCE FUNDAMENTALS

### OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** identify the components and explain the operation of a vehicle powertrain from an owner's perspective;
- B2.** identify and describe the components and service requirements of major vehicle systems, and related safety precautions, that an owner should be aware of;
- B3.** describe procedures for the proper care and maintenance of the interior and exterior of a vehicle;
- B4.** identify appropriate repair information and the tools and equipment required for safe basic service and maintenance of a vehicle.

### SPECIFIC EXPECTATIONS

#### B1. Powertrain Components

By the end of this course, students will:

- B1.1** describe the types of powerplants (*e.g., two-stroke, four-stroke, diesel, hybrid*) used in various vehicles;
- B1.2** identify the components of engines that require regular servicing (*e.g., oil and oil filter, air filter, spark plugs*);
- B1.3** explain, from an owner's perspective, the basic operation of various engine systems (*e.g., lubrication, cooling, starting, ignition, and fuel systems*) and the related safety and service considerations;
- B1.4** explain the operation of vehicle drivetrain components (*e.g., clutch, transmission, driveshaft, differential, axles, track*) and identify those that require regular servicing (*e.g., fluid change, lubrication*).

#### B2. Major Vehicle Systems

By the end of this course, students will:

- B2.1** locate and identify various components of vehicle systems that an owner should be aware of (*e.g., electrical system – battery; brakes – brake fluid reservoir; steering – tires; suspension – shocks*);
- B2.2** identify typical service requirements (*e.g., fluid change, parts replacement*) and intervals (*e.g., months, kilometres*) related to vehicle systems that an owner should be aware of;

- B2.3** identify the product information (*e.g., tire sizing*) and hazards and safety precautions (*e.g., battery explosion – wearing safety glasses*) that an owner should be aware of when servicing vehicle systems.

#### B3. Interior and Exterior Care

By the end of this course, students will:

- B3.1** describe, on the basis of research, interior and exterior vehicle cleaning and protection products and their applications;
- B3.2** identify various types and applications of polishes, waxes, and cleaners for vehicle finishes;
- B3.3** describe the pros and cons of various types of body corrosion prevention (*e.g., oil, dripless, electronic*) available through vehicle dealerships and aftermarket suppliers;
- B3.4** describe various techniques used to make minor repairs to a vehicle's finish (*e.g., stone-chip repair, small-dent repair*).

#### B4. Information, Tools, and Equipment for Basic Service and Maintenance

By the end of this course, students will:

- B4.1** consult the owner's manual as required for specific procedures, specifications, and products (*e.g., oils, fluids, fuses, bulbs*) related to the maintenance of a vehicle;

**B4.2** identify the tools and equipment required by a vehicle owner to perform basic service and maintenance procedures;

**B4.3** report on work in progress (*e.g., by completing a work order, parts list, and/or journal*), using terminology specific to vehicles (*e.g., acronyms, service terms*);

**B4.4** perform mathematical calculations related to vehicle maintenance and operation that are important from an owner's perspective (*e.g., calculate quantities, ratio of water to antifreeze, fuel consumption*), using appropriate resources (*e.g., owner's manual, service information*);

**B4.5** explain the benefits for an owner in keeping up-to-date service and maintenance records.

# C. VEHICLE MAINTENANCE SKILLS

## OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of service information and the importance of vehicle identification and information labels when performing vehicle maintenance and repair;
- C2.** perform basic engine and related system service following the owner's manual guidelines and appropriate service information sources, and identify system components that need to be professionally serviced;
- C3.** perform general vehicle maintenance and service as recommended in the owner's manual and outlined in appropriate service information sources, and identify system components that need to be professionally serviced.

## SPECIFIC EXPECTATIONS

### C1. Using Service Information

By the end of this course, students will:

- C1.1** identify and use appropriate service information sources (*e.g., shop manuals, online information, manufacturer's information*) as required for basic vehicle maintenance and repair;
- C1.2** identify the meaning of the letters and numbers of the vehicle identification number (VIN) (*e.g., place of origin, engine size, production year*) on the basis of their placement in the VIN;
- C1.3** locate and apply as required information found in the owner's manual and on the vehicle (*e.g., safety warnings, specifications such as tire size and recommended pressure, identification labels, graphics*) when performing basic maintenance and service procedures;
- C1.4** locate and correctly interpret graphics commonly found on vehicles and in owner's manuals that depict components being serviced (*e.g., jacking a vehicle, rotating tires, installing a serpentine belt*).

### C2. Engine Service

By the end of this course, students will:

- C2.1** locate the components of an engine lubrication system (*e.g., oil dipstick, oil filter, oil pan drain plug*) and safely and correctly service the system;

- C2.2** locate the components of an engine cooling system (*e.g., cooling fins, radiator cap, radiator hoses*) and safely and correctly service the system (*e.g., perform a freeze point test, system pressure test, boiling point test*);

- C2.3** locate the components of an engine starting system (*e.g., battery, battery cables, starting motor*) and safely and correctly service the system (*e.g., clean battery connections, use booster cables*);

- C2.4** locate the components of an ignition system (*e.g., spark plugs, ignition wires*) and identify those that need to be professionally serviced;

- C2.5** locate the components of an engine fuel system (*e.g., fuel cap, tank and lines, filter*) and identify those that need to be professionally serviced.

### C3. General Vehicle Maintenance and Service

By the end of this course, students will:

- C3.1** demonstrate the correct use of hand tools and equipment required for basic service and maintenance (*e.g., vehicle jacks, safety stands, wrenches*), store them safely, and maintain them in good working order;

- C3.2** locate the components of the electrical system (*e.g., battery, alternator, fuses, light bulbs*) and perform basic diagnostic and repair procedures (*e.g., check fuse, replace bulb, test and charge battery, install a trailer wiring harness*) safely and correctly;

- C3.3** locate the basic components of the brake system (*e.g., rotors, drums, friction material, brake fluid reservoir*) and identify those that need to be professionally serviced;
- C3.4** perform basic maintenance procedures related to the steering system (*e.g., basic wheel and tire service, changing a flat tire*) safely and correctly;
- C3.5** perform tasks related to a vehicle safety inspection (*e.g., check lights, horn, wipers, glass, door latches and locks*);
- C3.6** apply the proper procedures for maintaining the interior and exterior of a vehicle (*e.g., washing and waxing, upholstery cleaning and protection*);
- C3.7** prepare a vehicle for weather extremes (*e.g., winter cold, summer sun and heat*);
- C3.8** demonstrate an understanding of tasks necessary to prepare a vehicle for long-term and/or short-term storage (*e.g., add fuel system treatment, perform cylinder fogging, lubricate body hinges [on doors, hood, trunk or hatch]*).

# D. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

## OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** explain how vehicle ownership affects the environment and how vehicle owners can remedy or reduce harmful effects;
- D2.** explain how vehicle ownership and various aspects of the transportation industry affect society.

## SPECIFIC EXPECTATIONS

### D1. Technology and the Environment

By the end of this course, students will:

- D1.1** outline the legal requirements and environmental reasons for emission standards and for testing when required;
- D1.2** describe the options that vehicle owners have to choose environmentally friendly products (*e.g., biodegradable cleaners*) and procedures (*e.g., recycling of antifreeze*) in the repair and service of vehicles;
- D1.3** describe a vehicle owner's responsibilities with respect to recycling and/or disposing of waste products (*e.g., used oil, used batteries*) appropriately;
- D1.4** explain the importance of vehicle maintenance from an environmental perspective (*e.g., keeping tires properly inflated helps to maximize fuel efficiency and reduce emissions*).

### D2. Technology and Society

By the end of this course, students will:

- D2.1** analyse the safety features in today's vehicles (*e.g., electronic stability control, airbags, anti-lock brakes, roll-over protection, engine kill switch*) from a consumer's point of view;
- D2.2** describe the economic, environmental, and social effects that various aspects of the transportation industry have on a community (*e.g., economic: new businesses encouraged by good transportation links; environmental: pollution caused by exhaust emissions and road salting; social: community links expanded through personal vehicle use*);
- D2.3** assess from a consumer's point of view the pros and cons (*e.g., cost, availability, performance, reliability, emission levels*) of various types of fuel/energy sources used to power vehicles (*e.g., gasoline, propane, diesel, electrical/battery power, biodiesel, hybrid powerplant*).

## E. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

### OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** demonstrate an understanding of and compliance with occupational health and safety regulations and standards related to vehicle maintenance;
- E2.** identify and describe career opportunities in the transportation industry and the education and training required for these careers.

### SPECIFIC EXPECTATIONS

#### E1. Health and Safety

By the end of this course, students will:

- E1.1** identify and explain the importance of legislation and regulations related to procedures and operations used in transportation technology facilities (e.g., *Occupational Health and Safety Act [OHSA]*; *regulations and standards outlined in the Workplace Hazardous Materials Information System [WHMIS]*);
- E1.2** demonstrate good housekeeping and safety practices in the work environment (e.g., *cleaning up spills and leaks, keeping areas clean and clear of obstructions*);
- E1.3** use protective clothing and equipment (e.g., *safety glasses, hearing protection*) as required to ensure their own and others' safety in the work environment.

#### E2. Career Opportunities

By the end of this course, students will:

- E2.1** describe a variety of career opportunities in the transportation industry and the education and training required;
- E2.2** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the transportation industry (e.g., *mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);

- E2.3** demonstrate an understanding of and apply the Essential Skills that are important for success in the transportation industry, as identified in the Ontario Skills Passport (e.g., *finding information, measurement and calculation, decision making*);
- E2.4** demonstrate an understanding of and apply the work habits that are important for success in the transportation industry, as identified in the Ontario Skills Passport (e.g., *working safely, teamwork, reliability*);
- E2.5** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in transportation technology (e.g., *Passport to Safety certificate, skills checklist, Ontario Skills Passport Work Plan*), and explain why having a current portfolio is important for career development and advancement.